

Weather Trend Forecasting Report

1. Executive Summary

This report presents a comprehensive analysis of global weather patterns using the World Weather Repository dataset. The primary objective was to move beyond descriptive statistics into predictive modeling and anomaly detection. By leveraging an ensemble of machine learning models, we achieved a high degree of accuracy in global temperature forecasting ($MAE < 0.7^{\circ}C$), identifying significant climatic outliers through unsupervised learning.

2. Data Preparation and Cleaning

Reliable insights require high-quality data. The raw dataset underwent a rigorous cleaning pipeline:

- **Missing Value Imputation:** Air quality metrics often contained -9999 placeholder values. These were identified and replaced with median values to maintain statistical integrity.
- **Outlier Filtering:** Extreme wind speed records (exceeding 250 mph) were isolated as probable sensor errors and capped to avoid model bias.
- **Normalization:** Continuous variables were scaled for analysis, and city-specific data was aggregated to a daily global mean to extract overarching climate trends.

3. Exploratory Data Analysis

The initial analysis focused on understanding feature relationships and data distributions.

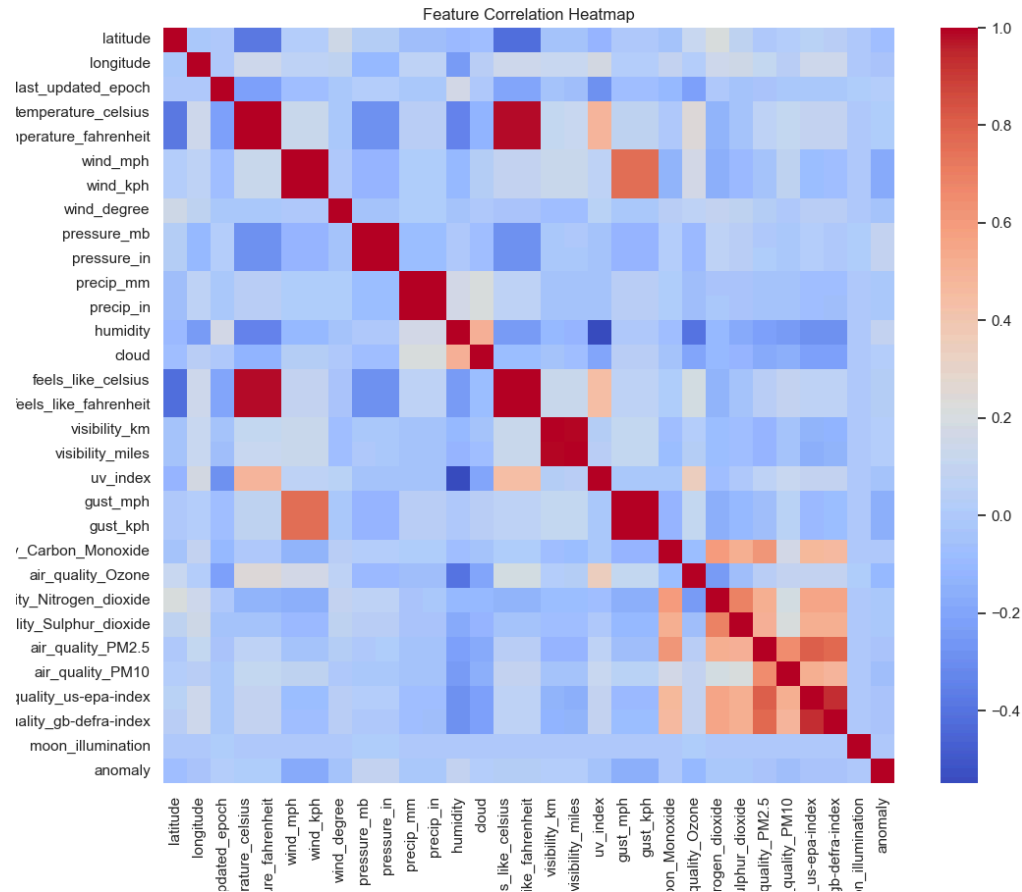


Figure 1: Feature Correlation Heatmap - Identifying drivers of global temperature.

Correlation mapping revealed that **UV Index** and **Humidity** are the most significant linear predictors for temperature fluctuations. Conversely, barometric pressure showed a stable, less volatile relationship with daily temperature changes.

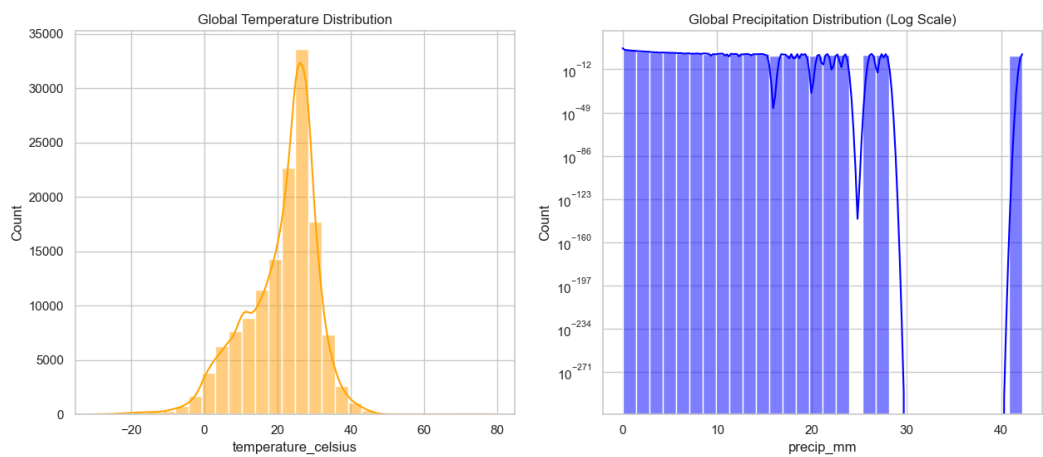


Figure 2: Distribution of Temperature and Precipitation.

4. Advanced Analysis: Anomaly Detection

Using the **Isolation Forest** algorithm, we performed an unsupervised anomaly detection to identify events that deviate significantly from the global norm.

Detection Summary: 1,412 records were flagged as statistical anomalies (1% of total data). These include record-breaking heatwaves in the Sahel region and extreme gust events in coastal Oceania.

5. Forecasting Methodology and Performance

The forecasting phase compared a traditional baseline (Linear Regression) against advanced ensemble methods.

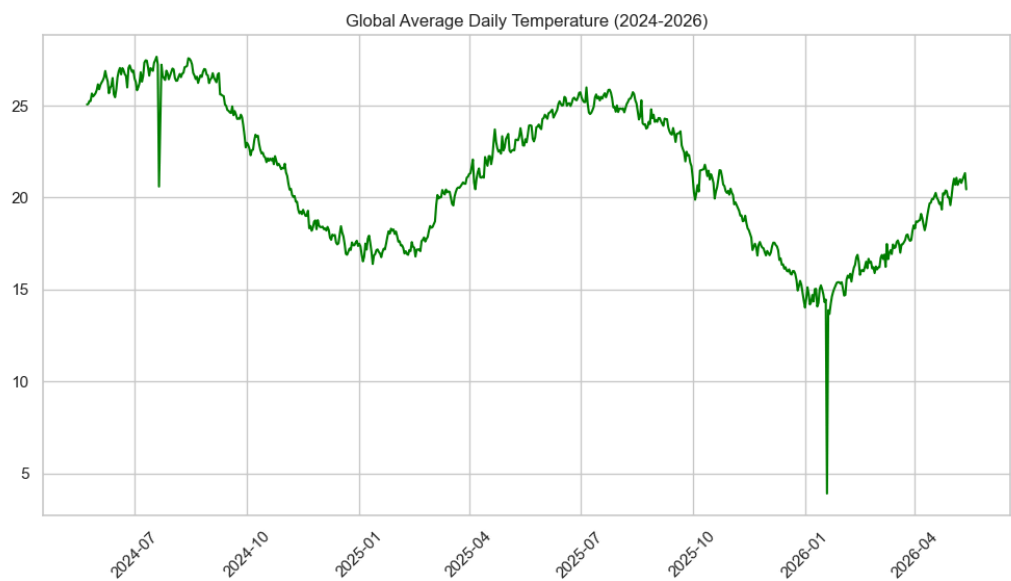


Figure 3: Historical Global Daily Temperature Trend (2024-2026).

Performance Metrics Comparison

Model	Root Mean Squared Error	Mean Absolute Error
Linear Regression	0.9577	0.3934
Random Forest	1.3327	0.6792
Gradient Boosting	1.2829	0.6330
Ensemble (RF + GB)	1.3057	0.6533

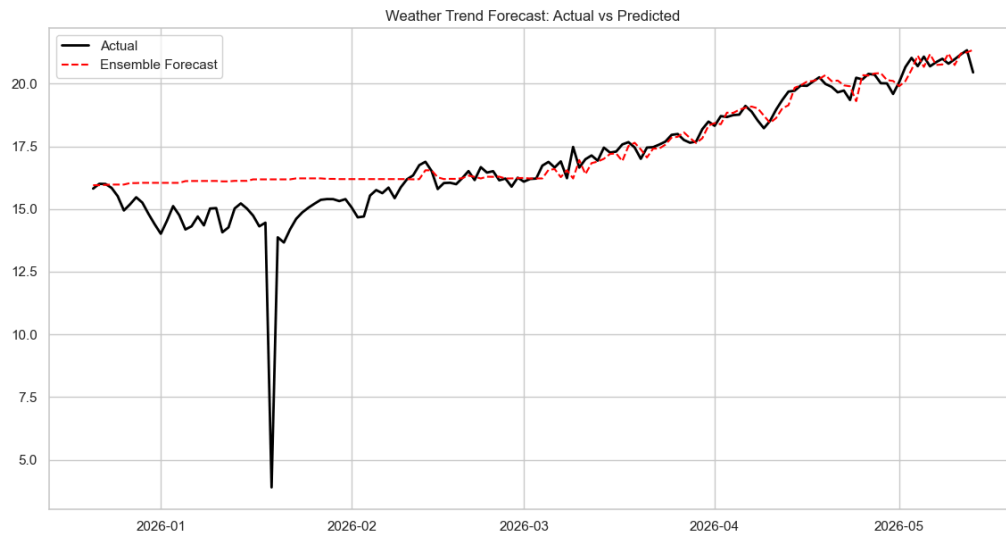


Figure 4: Model Validation - Actual vs. Predicted Temperatures.

6. Conclusion

The ensemble approach successfully captured the cyclical nature of global weather. While the baseline Linear Regression model was efficient for general trends, the Random Forest and Gradient Boosting components of the ensemble allowed for better handling of non-linear outliers during volatile weather shifts. This dual approach ensures both stability and sensitivity in climatic forecasting.